ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 7 Natural Sciences paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Accept other correct examples.

SECTION A

QUESTION 1

1 1.1 B ✓ 1.2 D ✓ 1.3 B ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 D ✓ 1.8 A ✓ 1.9 C ✓ 1.10 C ✓ (10)

QUESTION 2

2 2.1 H ✓ 2.2 C ✓ 2.3 E ✓ 2.4 I ✓ 2.5 G ✓ 2.6 A ✓ 2.7 D ✓ 2.8 J ✓ 2.9 F ✓ 2.10 B ✓ (10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3

3.1 Living things are found in the soil and on land (lithosphere), in water (hydrosphere) and in the air (atmosphere) ✓, so the biosphere is made up of parts of all three ✓. (2)

3.2.1 Fern ✓, pine tree ✓, moss ✓ (3)

3.2.2 Any TWO differences, two marks each: (4)
Monocots have one seed leaf (cotyledon) ✓; dicots have two ✓.
Monocots have parallel leaf veins ✓; dicots have net-like veins ✓.
Monocots have flower parts in threes; dicots in fours or fives.
Monocots have fibrous roots; dicots usually have a taproot.

3.3 Any TWO: a spider has 8 legs, an insect has 6 ✓; a spider has 2 body parts, an insect has 3 ✓; insects have antennae, spiders do not; many insects have wings, spiders do not (2)

3.4 (a) The monthly loss of blood and the lining of the uterus through the vagina ✓ when an egg has not been fertilised ✓ (3)
(b) Testes (testicles) ✓

3.5 (a) Any TWO: height ✓; eye colour ✓; skin colour; hair type; shoe size; blood group (4)
(b) If the environment changes (e.g. a new disease or drought), some individuals have features that help them survive ✓, so the species does not die out ✓.

3.6 Any TWO: water (moisture) ✓; warmth (suitable temperature) ✓; oxygen (air) (2)

[20]

QUESTION 4

4.1.1 Copper ✓ – a good conductor of electricity (and ductile) ✓ (2)

4.1.2 Glass ✓ – transparent (lets light through) ✓ (2)

4.1.3 Rubber ✓ – flexible / waterproof / grips the road ✓ (2)

4.2 Any TWO: saves raw materials (natural resources) ✓; less waste goes to landfill sites ✓; saves energy; reduces pollution; creates jobs (2)

4.3 Heat the salt water until it boils ✓. The water evaporates to form water vapour (steam), but the salt stays behind ✓. The vapour passes through a cool condenser ✓ where it condenses back into liquid water, which is collected as pure (distilled) water ✓. (4)

4.4 (a) Any TWO: vinegar ✓; lemon juice ✓; cold drinks; tomato juice; sour milk (5)
(b) Any TWO: soap ✓; toothpaste ✓; bicarbonate of soda; drain cleaner; milk of magnesia; ammonia cleaners
(c) Pink / red ✓

4.5 (a) Metals: magnesium and aluminium ✓ Non-metals: sulfur and carbon ✓ (3)
(b) Mg ✓

[20]

QUESTION 5

5.1 At the start the ball has gravitational potential energy ✓. As it falls, the potential energy changes into kinetic energy ✓ (it speeds up). When it hits the ground, the energy is changed into sound and heat (and some elastic energy if it bounces) ✓. (3)

5.2 (a) At the top of the highest hill ✓ (b) At the lowest point (bottom of the biggest drop) ✓ (2)

5.3.1 The type of covering material ✓ (1)

5.3.2 80 − 60 = 20 °C ✓ (1)

5.3.3 Wool ✓ – its temperature dropped the least (only 12 °C), so it lost heat the slowest ✓ (2)

5.3.4 Any TWO: volume of water ✓; starting temperature ✓; size/type of tin; time; room temperature (2)

5.4 Any TWO sources ✓✓ with ONE advantage each ✓✓: (4)
Solar – free, will not run out, no pollution; Wind – no pollution, free once built; Hydro-electric – no pollution, can produce large amounts

5.5 (a) Less energy is lost as heat in the power lines ✓ (3)
(b) It lowers (steps down) the voltage ✓ to a safe level (230 V) for homes ✓

5.6 Any TWO: switch off lights and appliances not in use ✓; use LED / energy-saving bulbs ✓; use a solar geyser or geyser blanket; boil only the water you need; dry clothes in the sun (2)

[20]

QUESTION 6

6.1 Sun, Earth and Moon in a straight line ✓ with the Earth between the Sun and the Moon ✓; all three labelled ✓ (3)

6.2 The Earth moves between the Sun and the Moon ✓ and the Earth's shadow falls on the Moon, so the Moon becomes dark (often reddish) ✓. (2)

6.3 At new moon and full moon, when the Sun, Earth and Moon are in a line ✓. The high tides are higher and the low tides are lower than usual ✓. (2)

6.4 Any TWO: summer days are longer, winter days shorter ✓; the Sun is higher in the sky in summer ✓; summer is warmer, winter colder (2)

6.5 The Earth spins (rotates) on its axis ✓; the side facing the Sun has day while the side facing away has night ✓. (2)

6.6 Any TWO: as a calendar to count months ✓; to know when to plant and harvest crops ✓; to find direction when travelling (navigation); for stories and ceremonies (2)

6.7 Any TWO: dark skies far from city lights ✓; dry, clear air with few clouds ✓; high above sea level (2)

6.8 The Sun always lights up half of the Moon ✓. As the Moon orbits the Earth ✓, we see different amounts of the lit half from Earth, so it seems to change shape ✓. (3)

6.9 The Earth takes about 365¼ days to orbit the Sun ✓; the four quarter days add up to one extra day every four years ✓. (2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF MARKING GUIDELINE